

Consistent Partial Least Squares: A cSEM tutorial

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Abstract

This tutorial article provides a comprehensive, step-by-step illustration of consistent partial least squares (PLSc) using the **cSEM** package in R. PLSc is a variant of partial least squares path modeling, a composite-based approach to structural equation modeling, that produces consistent parameter estimates for latent variable models. In particular, PLSc corrects construct correlations for attenuation using Dijkstra-Henseler's composite reliability coefficient ϱ_A . Building on the employee retention example introduced by Cheah, Sarstedt, Hair, and Ringle (2026), we demonstrate how to install and load **cSEM**, specify the model using **lavaan**-style syntax, estimate the PLSc-SEM model, evaluate measurement model quality (internal consistency reliability, convergent validity, and discriminant validity), and assess the structural model (collinearity, effect sizes, model fit, and path coefficients with bootstrap inference). We further contrast the PLSc-SEM results with standard PLS-SEM estimates and discuss practical considerations for using PLSc-SEM in empirical research.

Keywords: measurement error, bias correction, partial least squares structural equation modeling (PLS-SEM), factor score regression

1 Introduction

Structural equation modeling (SEM) is a versatile multivariate analysis framework used in the behavioral, social, medical, and management sciences to study complex relationships between constructs and their relations with observed variables (e.g., Kline, 2023; Marcoulides & Schumacker, 1996). To estimate structural equation models, various approaches have been proposed, including maximum-likelihood (Jöreskog, 1970), weighted least squares (Browne, 1974), factor score regression (Devlieger & Rosseel, 2017; Skrandal & Laake, 2001) and (integrated) generalized structured component analysis (Hwang et al., 2021; Hwang & Takane, 2004)

A composite-based estimator to SEM that gained popularity over the last decades in various fields of social sciences (Cook & Forzani, 2023), including marketing (Sarstedt et al., 2022), information systems research (Benitez, Henseler, Castillo, & Schuberth, 2020; Evermann & Rönkkö, 2023), human resource management (Ringle, Sarstedt, Mitchell, & Gudergan, 2020), and tourism research (Müller, Schuberth, & Henseler, 2018), is partial least squares path modeling (PLS-PM, Wold, 1982), also known as partial least squares structural equation modeling (PLS-SEM, Hair, Ringle, & Sarstedt, 2011). PLS-PM is a two-step estimation procedure (Schuberth, Zaza, & Henseler, 2023). In the first step, construct scores are obtained by the partial least squares algorithm. Subsequently, these construct scores are used to estimate the model parameters. Although PLS-PM is computationally efficient, it is known that in its original form to produce inconsistent estimates for models involving latent variables due to attenuation bias (e.g., Dijkstra, 1985; Hui & Wold, 1982; Rönkkö & Evermann, 2013; Schuberth, Rosseel, et al., 2023).

To address this limitation, consistent partial least squares (PLSc, Dijkstra, 2013; Dijkstra & Henseler, 2015a, 2015b; Dijkstra & Schermelleh-Engel, 2014) was introduced. PLSc is based on PLS-PM, but applies a correction for attenuation to obtain consistent estimates of reflective measurement models and path coefficient between latent variables. Additionally, PLSc has been extended to deal with ordinal variables (Schuberth, Henseler, & Dijkstra, 2018), randomly occurring outliers (Schamberger, Schuberth, Henseler, & Dijkstra, 2020) and multicollinearity issues (Jung & Park, 2018).

To apply PLS-PM and PLSc, various implementations have been developed (Venturini, Mehmetoglu, & Latan, 2023). They can be distinguished into: (i) proprietary software and (ii) open-source software. Proprietary software for conducting PLS-PM and PLSc includes SmartPLS (Ringle, Wende, & Becker, 2024), WarpPLS (Kock, 2025) and ADANCO (Henseler, 2025). On the other hand, open-source software alternatives include implementations in the statistical programming environment R (R Core Team, 2025) such as *matrixpls* (Rönkkö, 2022), *SEM-inR* (Ray, Danks, & Calero Valdez, 2026), *cSEM* (Rademaker & Schuberth, 2020) and *plspm* (Sanchez, Trinchera, Russolillo, & Bertrand, 2025). Similarly, it includes implementations in python (Python Software Foundation, 2024) such as *plspm* (Humble, 2024). While proprietary software such as SmartPLS often shows high ease of use (Cheah, Magno, & Cassia, 2024), they conflict with the open science principles (UNESCO, 2021). For example, their codebase is not openly accessible of transparency (Morin et al., 2012), reproducibility (Ince, Hatton, & Graham-Cumming, 2012), and accessibility/inclusiveness (). Open-source software overcomes these limitations (Barba, 2022). Therefore, this tutorial presents the R package *cSEM* as a full-fledged open-source alternative to SmartPLS to apply PLS-PM and PLSc.

Barba (2022)

(Chuah et al., 2021)

The remainder of the tutorial is structured as follows. Section Section Section

2 cSEM R Illustration

2.1 Getting started with R

To use the cSEM package, users first need to install R and an integrated development environment (IDE) such as RStudio. For the installation, we refer to the official guides for R (<https://cran.r-project.org>) and RStudio (<https://posit.co/download/rstudio-desktop>).

When RStudio is opened for the first time, it displays four panes (Figure 1). The two most important ones are the *script editor* (upper left), where code is written and saved, and the *Console* (lower left), where the code is executed and results are printed. The *Environment* pane (upper right) lists the objects currently loaded into memory, and the fourth pane (lower right) provides access to files, plots, and help pages.

Although code can be typed directly into the console, it is good practice to work in a script, since a script can be saved, shared, and re-run, which is essential for reproducible analyses. To create a new script, click **File** → **New File** → **R Script**. The typical workflow is to write code in the editor and run the selected line(s) with **Ctrl + Enter**. Any line beginning with **#** is treated as a comment: it documents what the code does and is ignored when the code runs.

```
# This is a comment and is not executed.  
1 + 1 # Code is executed; the result is printed in the Console.  
  
[1] 2
```

Before cSEM can be used, it must be installed. The cSEM package is available on CRAN, so the most recent stable version can be installed with:

```
install.packages("cSEM")
```

Installation only needs to be done once per machine (rerun this line to update the package later). Moreover, development versions can be installed from GitHub: <https://github.com/FloSchuberth/cSEM>. After installation, the package must be loaded at the start of every new R session before its functions become available:

```
library(cSEM)
```

```
Attaching package: 'cSEM'  
The following object is masked from 'package:stats':  
    predict  
The following object is masked from 'package:grDevices':  
    savePlot
```

With cSEM installed and loaded, we can now import the data and specify the model, as described in the following sections.

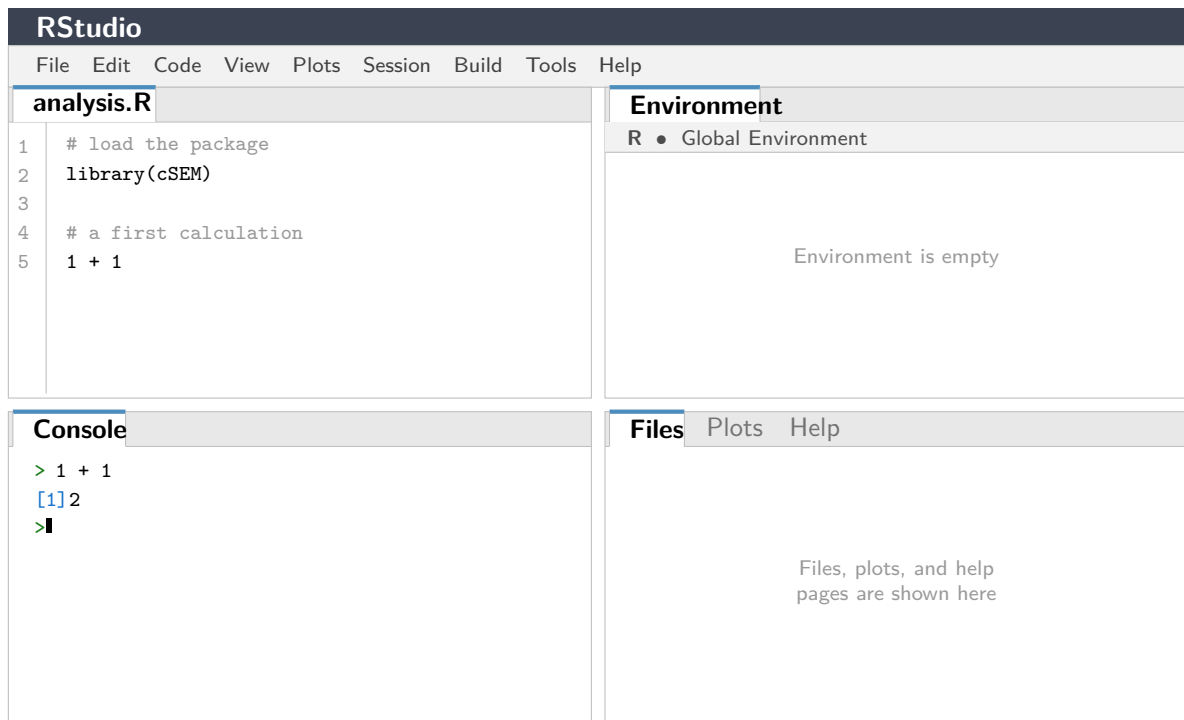


Figure 1: The four panes of the RStudio interface: the script editor (upper left), the *Console* (lower left), the *Environment* pane (upper right), and the files/plots/help pane (lower right).

2.2 Model and Data

The *cSEM* package (Rademaker & Schubert, 2020) offers a variety of composite-based approaches to SEM including PLS-PM and PLSc. To illustrate the application of PLSc using *cSEM*, we use the data and model used in Cheah et al. (2026) to demonstrate SmartPLS. Similarly, we compare the results to those of the maximum-likelihood estimation as implemented in the R package *lavaan* (Rosseel, 2012; Rosseel, Jorgensen, & De Wilde, 2025).

Following Cheah et al. (2026), our illustration draws on the employee retention model introduced by Hair, Black, Babin, and Anderson (2019, Chapter 13). The model explains the effects of organizational commitment (OC) and job satisfaction (JS) on employees' staying intentions (SI), with work environment perceptions (EP) and attitudes toward coworkers (AC) as exogenous antecedent constructs. Five constructs are reflectively measured by a total of 21 indicators. A detailed description of the indicators can be found in Cheah et al. (2026). The model is presented in Figure 2.

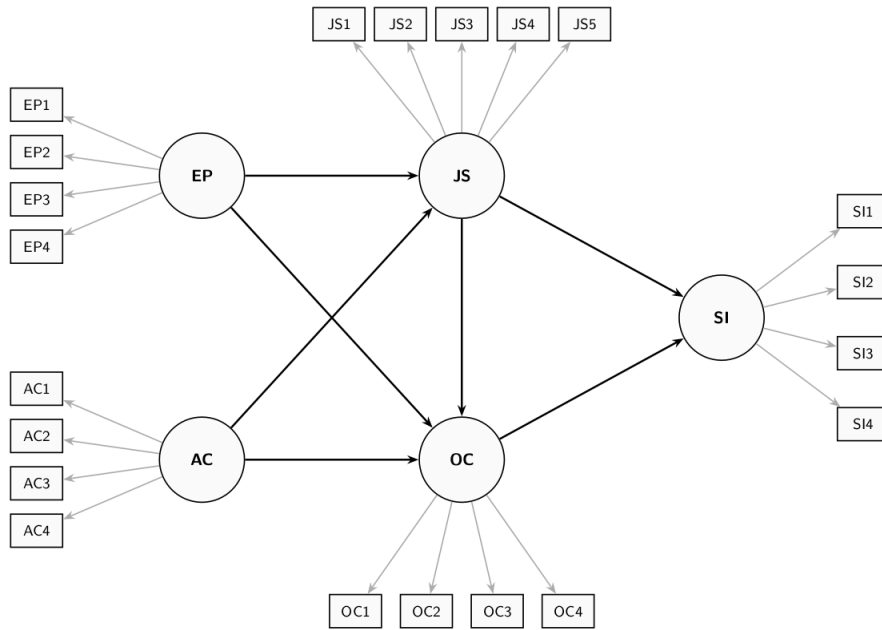


Figure 2: Employee Retention Model

The dataset used for model estimation is publicly available from the SmartPLS website: <https://www.smartpls.com/documentation/sample-projects/employee-retention>. The synthetic dataset comprises $N = 400$ observations from HBAT Industries and was generated to satisfy the assumptions of factor-based SEM, including multivariate normality. For this tutorial, we assume the data have been downloaded as a CSV file. After downloading the dataset, it needs to be loaded into the R Environment. The Employee Retention dataset is stored as an CSV File using ';' as the separator, thus it can be loaded into R using following command:

```
# Import dataset
HBAT_SEM <- read.csv("data/HBAT_SEM.csv", sep=";")

# Select relevant variables
HBAT_SEM_rel <- HBAT_SEM[, c("AC1", "AC2", "AC3", "AC4",
                             "EP1", "EP2", "EP3", "EP4",
                             "JS1", "JS2", "JS3", "JS4", "JS5",
                             "OC1", "OC2", "OC3", "OC4",
                             "SI1", "SI2", "SI3", "SI4")]

# show dimensions of the relevant dataset
dim(HBAT_SEM_rel)

[1] 400 21

# show some basic statistics for all variables in the dataset
summary(HBAT_SEM_rel)
```

AC1	AC2	AC3	AC4
Min. :1.00	Min. :1.000	Min. :1.000	Min. :1.000
1st Qu.:1.75	1st Qu.:2.000	1st Qu.:1.000	1st Qu.:2.000
Median :3.00	Median :4.000	Median :3.000	Median :3.000
Mean :2.76	Mean :3.553	Mean :2.775	Mean :3.211
3rd Qu.:4.00	3rd Qu.:5.000	3rd Qu.:4.000	3rd Qu.:4.000

Max. :5.00	Max. :6.000	Max. :5.000	Max. :6.000
			NA's :1
EP1	EP2	EP3	EP4
Min. : 0.000	Min. : 0.000	Min. : 0.000	Min. :1.00
1st Qu.: 8.000	1st Qu.: 8.000	1st Qu.: 8.000	1st Qu.:5.00
Median : 9.000	Median : 9.000	Median : 9.000	Median :6.00
Mean : 8.527	Mean : 8.848	Mean : 8.928	Mean :5.83
3rd Qu.:10.000	3rd Qu.:10.000	3rd Qu.:10.000	3rd Qu.:7.00
Max. :10.000	Max. :10.000	Max. :10.000	Max. :7.00
			NA's :1
JS1	JS2	JS3	JS4
Min. :1.000	Min. :1.000	Min. :1.000	Min. :1.000
1st Qu.:3.000	1st Qu.:3.000	1st Qu.:2.000	1st Qu.:2.000
Median :4.000	Median :4.000	Median :3.000	Median :3.000
Mean :4.197	Mean :4.202	Mean :3.215	Mean :2.667
3rd Qu.:5.000	3rd Qu.:5.000	3rd Qu.:4.000	3rd Qu.:4.000
Max. :7.000	Max. :7.000	Max. :6.000	Max. :5.000
JS5	OC1	OC2	OC3
Min. : 0.00	Min. : 0.000	Min. : 0.000	Min. : 0.000
1st Qu.: 39.97	1st Qu.: 3.000	1st Qu.: 8.000	1st Qu.: 8.000
Median : 55.35	Median : 5.000	Median : 9.000	Median : 9.000
Mean : 54.83	Mean : 4.888	Mean : 8.418	Mean : 8.655
3rd Qu.: 70.44	3rd Qu.: 7.000	3rd Qu.:10.000	3rd Qu.:10.000
Max. :100.00	Max. :10.000	Max. :10.000	Max. :10.000
OC4	SI1	SI2	SI3
Min. : 0.000	Min. :1.000	Min. :1.000	Min. :1.000
1st Qu.: 8.000	1st Qu.:4.000	1st Qu.:4.000	1st Qu.:3.000
Median : 9.000	Median :4.000	Median :4.000	Median :3.000
Mean : 8.405	Mean :4.202	Mean :4.205	Mean :3.473
3rd Qu.:10.000	3rd Qu.:5.000	3rd Qu.:5.000	3rd Qu.:4.000
Max. :10.000	Max. :5.000	Max. :5.000	Max. :5.000
SI4			
Min. :1.00			
1st Qu.:3.00			
Median :4.00			
Mean :3.48			
3rd Qu.:4.00			
Max. :5.00			

In the summary, we see that we have missing values in the variables AC4 and EP4. Before we can continue we need to use imputation methods or delete incomplete observations. Deleting incomplete observations can be done via following command:

```
# filter the dataset, only select complete cases
HBAT_SEM_rel <- HBAT_SEM_rel[complete.cases(HBAT_SEM_rel),]

# the number of rows changed from 400 to 398
dim(HBAT_SEM_rel)

[1] 398 21
```

When using their own data, users must ensure it is stored as a data frame or a matrix before passing it to `cSEM` functions; if necessary, the data can be converted with `as.data.frame()` or `as.matrix()`.

2.3 Model specification

In `cSEM`, models are specified using the `lavaan` model syntax. The `lavaan` package does not need to be installed to use the syntax. In particular, the `=~` and the `<~` operators are used to define the relations between the constructs and their indicators. Specifically, the `=~` operator is used to specify reflective measurement models, while the `<~` is used to specify composite models (see , for an explanation of reflective and composite models). In addition, the `~` operator is used to specify the relations among constructs. In case, an observed variable should be directly accounted for in the structural model, it must be specified as a single indicator construct, as common in PLS-PM (). The `cSEM` specification of the retention model illustrated in Figure 2 is as follows:

```
model <- "  
  # Measurement model  
  AC =~ AC1 + AC2 + AC3 + AC4  
  EP =~ EP1 + EP2 + EP3 + EP4  
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5  
  OC =~ OC1 + OC2 + OC3 + OC4  
  SI =~ SI1 + SI2 + SI3 + SI4  
  
  # Structural model  
  JS ~ EP + AC  
  OC ~ EP + AC + JS  
  SI ~ JS + OC  
"
```

3 Estimating the model with `cSEM`

3.1 Running the estimation

The central function is `csem()`. The minimal input required for the `csem()` function is a dataset and a model:

- .data** A data frame or matrix with column names matching the indicator names used in the model description. Possible column types of data proved are: "logical", "numeric" ("double" or "integer"), "factor" ("ordered" and/or "unordered"), "character" or a mix of several types.
- .model** A model string written in `lavaan` syntax, with `=~` for measurement relations and `~` for structural (regression) relations.

By default weights are estimated using the partial least squares path modeling algorithm ("PLS-PM").

To ultimately estimate with the `csem` function, we call it and save the results as `res_csem`:


```
# PLSc-SEM estimation
res_csem <- csem(
  .data = HBAT_SEM_rel,
  .model = model
)
```

```
# Verify convergence
verify(res_csem)
```

```
-----
Verify admissibility:
```

```
  admissible
```

```
Details:
```

Code	Status	Description
1	ok	Convergence achieved
2	ok	All absolute standardized loading estimates <= 1
3	ok	Construct VCV is positive semi-definite
4	ok	All reliability estimates <= 1
5	ok	Model-implied indicator VCV is positive semi-definite

```
-----
```

The `verify()` function checks whether the PLS algorithm converged, whether all construct reliability estimates (ρ_A) are positive (a necessary condition for the PLSc correction), and whether the model-implied indicator correlation matrix is admissible. If all checks pass, the estimation can be considered successful.

To obtain a comprehensive overview of all results, including loadings, path coefficients, reliability measures, and R^2 values, use:

```
# Full results summary
summary_res <- summarize(res_csem)
print(summary_res)
```

```
-----
Overview -----
```

```
General information:
```

```
-----
Estimation status           = Ok
Number of observations      = 398
Weight estimator           = PLS-PM
Inner weighting scheme     = "path"
Type of indicator correlation = Pearson
Path model estimator       = OLS
Second-order approach      = NA
Type of path model         = Linear
Disattenuated              = Yes (PLSc)
```

```
Construct details:
```

Name	Modeled as	Order	Mode
EP	Common factor	First order	"modeA"
AC	Common factor	First order	"modeA"
JS	Common factor	First order	"modeA"
OC	Common factor	First order	"modeA"
SI	Common factor	First order	"modeA"

----- Estimates -----

Estimated path coefficients:

=====

Path	Estimate	Std. error	t-stat.	p-value
JS ~ EP	0.2448	NA	NA	NA
JS ~ AC	-0.0033	NA	NA	NA
OC ~ EP	0.4288	NA	NA	NA
OC ~ AC	0.1726	NA	NA	NA
OC ~ JS	0.0961	NA	NA	NA
SI ~ JS	0.1312	NA	NA	NA
SI ~ OC	0.4900	NA	NA	NA

Estimated loadings:

=====

Loading	Estimate	Std. error	t-stat.	p-value
EP =~ EP1	0.6341	NA	NA	NA
EP =~ EP2	0.8787	NA	NA	NA
EP =~ EP3	0.7678	NA	NA	NA
EP =~ EP4	0.8231	NA	NA	NA
AC =~ AC1	0.7691	NA	NA	NA
AC =~ AC2	0.6765	NA	NA	NA
AC =~ AC3	0.8216	NA	NA	NA
AC =~ AC4	0.9934	NA	NA	NA
JS =~ JS1	0.6248	NA	NA	NA
JS =~ JS2	0.6965	NA	NA	NA
JS =~ JS3	0.7182	NA	NA	NA
JS =~ JS4	0.6319	NA	NA	NA
JS =~ JS5	0.9004	NA	NA	NA
OC =~ OC1	0.4252	NA	NA	NA
OC =~ OC2	0.9575	NA	NA	NA
OC =~ OC3	0.6859	NA	NA	NA
OC =~ OC4	0.8566	NA	NA	NA
SI =~ SI1	0.8170	NA	NA	NA
SI =~ SI2	0.8705	NA	NA	NA
SI =~ SI3	0.6778	NA	NA	NA
SI =~ SI4	0.8959	NA	NA	NA

Estimated weights:

=====

Weight	Estimate	Std. error	t-stat.	p-value
EP <~ EP1	0.2425	NA	NA	NA
EP <~ EP2	0.3361	NA	NA	NA
EP <~ EP3	0.2937	NA	NA	NA
EP <~ EP4	0.3148	NA	NA	NA

AC <~ AC1	0.2707	NA	NA	NA
AC <~ AC2	0.2381	NA	NA	NA
AC <~ AC3	0.2892	NA	NA	NA
AC <~ AC4	0.3496	NA	NA	NA
JS <~ JS1	0.2223	NA	NA	NA
JS <~ JS2	0.2478	NA	NA	NA
JS <~ JS3	0.2556	NA	NA	NA
JS <~ JS4	0.2249	NA	NA	NA
JS <~ JS5	0.3204	NA	NA	NA
OC <~ OC1	0.1741	NA	NA	NA
OC <~ OC2	0.3920	NA	NA	NA
OC <~ OC3	0.2808	NA	NA	NA
OC <~ OC4	0.3507	NA	NA	NA
SI <~ SI1	0.2882	NA	NA	NA
SI <~ SI2	0.3071	NA	NA	NA
SI <~ SI3	0.2391	NA	NA	NA
SI <~ SI4	0.3161	NA	NA	NA

Estimated construct correlations:

=====

Correlation	Estimate	Std. error	t-stat.	p-value
EP ~ AC	0.2549	NA	NA	NA

----- Effects -----

Estimated total effects:

=====

Total effect	Estimate	Std. error	t-stat.	p-value
JS ~ EP	0.2448	NA	NA	NA
JS ~ AC	-0.0033	NA	NA	NA
OC ~ EP	0.4523	NA	NA	NA
OC ~ AC	0.1722	NA	NA	NA
OC ~ JS	0.0961	NA	NA	NA
SI ~ EP	0.2538	NA	NA	NA
SI ~ AC	0.0840	NA	NA	NA
SI ~ JS	0.1783	NA	NA	NA
SI ~ OC	0.4900	NA	NA	NA

Estimated indirect effects:

=====

Indirect effect	Estimate	Std. error	t-stat.	p-value
OC ~ EP	0.0235	NA	NA	NA
OC ~ AC	-0.0003	NA	NA	NA
SI ~ EP	0.2538	NA	NA	NA
SI ~ AC	0.0840	NA	NA	NA
SI ~ JS	0.0471	NA	NA	NA

For a more compact overview

summary_res\$Estimates>Loading_estimates

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP =~ EP1	Common factor	0.6341011		NA	NA
2	EP =~ EP2	Common factor	0.8786819		NA	NA

```

3 EP =~ EP3 Common factor 0.7678021 NA NA NA
4 EP =~ EP4 Common factor 0.8230691 NA NA NA
5 AC =~ AC1 Common factor 0.7691299 NA NA NA
6 AC =~ AC2 Common factor 0.6765259 NA NA NA
7 AC =~ AC3 Common factor 0.8215860 NA NA NA
8 AC =~ AC4 Common factor 0.9933858 NA NA NA
9 JS =~ JS1 Common factor 0.6248337 NA NA NA
10 JS =~ JS2 Common factor 0.6964587 NA NA NA
11 JS =~ JS3 Common factor 0.7182274 NA NA NA
12 JS =~ JS4 Common factor 0.6319045 NA NA NA
13 JS =~ JS5 Common factor 0.9004003 NA NA NA
14 OC =~ OC1 Common factor 0.4252105 NA NA NA
15 OC =~ OC2 Common factor 0.9574682 NA NA NA
16 OC =~ OC3 Common factor 0.6858765 NA NA NA
17 OC =~ OC4 Common factor 0.8565670 NA NA NA
18 SI =~ SI1 Common factor 0.8170099 NA NA NA
19 SI =~ SI2 Common factor 0.8705194 NA NA NA
20 SI =~ SI3 Common factor 0.6778349 NA NA NA
21 SI =~ SI4 Common factor 0.8959341 NA NA NA

```

```
summary_res$Estimates$Path_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	JS ~ EP	Common factor	0.244824960	NA	NA	NA
2	JS ~ AC	Common factor	-0.003282073	NA	NA	NA
3	OC ~ EP	Common factor	0.428781485	NA	NA	NA
4	OC ~ AC	Common factor	0.172554413	NA	NA	NA
5	OC ~ JS	Common factor	0.096069429	NA	NA	NA
6	SI ~ JS	Common factor	0.131226588	NA	NA	NA
7	SI ~ OC	Common factor	0.490013776	NA	NA	NA

3.2 Assessing the measurement model

The next step involves assessing the measurement model results, according the guidelines described in !!!Benites!!!. The `assess()` function in `cSEM` provides all relevant quality criteria:

```
# Compute all quality criteria
quality <- assess(res_csem)
```

```
Warning: The following warning occurred in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
```

```
Warning: The following warning occurred in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
```

```
Warning in log(x[[3]]): NaNs produced
```

```
# Internal consistency reliability
quality
```

Construct	AVE	R2	R2_adj
-----------	-----	----	--------

EP	0.6103	NA	NA
AC	0.6778	NA	NA
JS	0.5203	0.0595	0.0548
OC	0.5754	0.2826	0.2772
SI	0.6719	0.2845	0.2808

----- Common (internal consistency) reliability estimates -----

Construct	Cronbachs_alpha	Joereskogs_rho	Dijkstra-Henselers_rho_A
EP	0.8596	0.8607	0.8717
AC	0.8936	0.8919	0.9105
JS	0.8450	0.8417	0.8568
OC	0.8328	0.8344	0.8879
SI	0.8890	0.8902	0.8989

----- Alternative (internal consistency) reliability estimates -----

Construct	RhoC	RhoC_mm	RhoC_weighted
EP	0.8607	0.8557	0.8717
AC	0.8919	0.8765	0.9105
JS	0.8417	0.8267	0.8568
OC	0.8344	0.8029	0.8879
SI	0.8902	0.8860	0.8989

Construct	RhoC_weighted_mm	RhoT	RhoT_weighted
EP	0.8717	0.8596	0.8522
AC	0.9105	0.8936	0.8923
JS	0.8568	0.8450	0.8427
OC	0.8879	0.8328	0.7954
SI	0.8989	0.8890	0.8830

----- Distance and fit measures -----

Geodesic distance = 0.3252628
 Squared Euclidean distance = 1.150488
 ML distance = 1.89721

Chi_square = 753.1923
 Chi_square_df = 4.161284
 CFI = 0.8647056
 CN = 113.476
 GFI = 0.7555901
 IFI = 0.865627
 NFI = 0.830333
 NNFI = 0.8430286
 RMSEA = 0.08923526
 RMS_theta = 0.04595267
 SRMR = 0.07057243

Degrees of freedom = 181

----- Model selection criteria -----

Construct	AIC	AICc	AICu
JS	-19.4331	380.6687	-16.4217
OC	-125.2111	274.9420	-121.1909
SI	-128.2165	271.8852	-125.2052

Construct	BIC	FPE	GM
JS	-7.4737	0.9523	415.1565
OC	-109.2653	0.7301	420.5491
SI	-116.2572	0.7246	503.0830

Construct	HQ	HQc	Mallows_Cp
JS	-14.6961	-14.5595	5.1971
OC	-118.8951	-118.6760	6.6033
SI	-123.4796	-123.3430	93.1236

----- Variance inflation factors (VIFs) -----

Dependent construct: 'JS'

Independent construct	VIF value
EP	1.0695
AC	1.0695

Dependent construct: 'OC'

Independent construct	VIF value
EP	1.1333
AC	1.0695
JS	1.0633

Dependent construct: 'SI'

Independent construct	VIF value
JS	1.0465
OC	1.0465

----- Effect sizes (Cohen's f^2) -----

Dependent construct: 'JS'

Independent construct	f^2
EP	0.0596
AC	0.0000

Dependent construct: 'OC'

Independent construct	f^2
EP	0.2262
AC	0.0388
JS	0.0121

Dependent construct: 'SI'

Independent construct	f ²
JS	0.0230
OC	0.3206

----- Discriminant validity assessment -----

Heterotrait-monotrait ratio of correlations matrix (HTMT matrix)

	EP	AC	JS	OC	SI
EP	1.0000000	0.0000000	0.0000000	0.0000000	0
AC	0.2555939	1.0000000	0.0000000	0.0000000	0
JS	0.2437979	0.05239146	1.0000000	0.0000000	0
OC	0.4939933	0.27449892	0.2086788	1.0000000	0
SI	0.5672796	0.30936763	0.2313363	0.5005239	1

Advanced heterotrait-monotrait ratio of correlations matrix (HTMT2 matrix)

	EP	AC	JS	OC	SI
EP	1.0000000	0.0000000	0.0000000	0.0000000	0
AC	0.2529206	1.0000000	0.0000000	0.0000000	0
JS	0.2376530	NaN	1.0000000	0.0000000	0
OC	0.4712082	0.2514075	0.1977162	1.0000000	0
SI	0.5661731	0.3083707	0.2219788	0.4705377	1

Fornell-Larcker matrix

	EP	AC	JS	OC	SI
EP	0.61028217	0.064998129	0.059530244	0.24622791	0.31507449
AC	0.06499813	0.677766756	0.003496994	0.08268627	0.09452102
JS	0.05953024	0.003496994	0.520269313	0.04447507	0.05502128
OC	0.24622791	0.082686268	0.044475067	0.57542078	0.26800117
SI	0.31507449	0.094521019	0.055021276	0.26800117	0.67186684

----- Effects -----

Estimated total effects:

=====

Total effect	Estimate	Std. error	t-stat.	p-value
JS ~ EP	0.2448	NA	NA	NA
JS ~ AC	-0.0033	NA	NA	NA
OC ~ EP	0.4523	NA	NA	NA
OC ~ AC	0.1722	NA	NA	NA
OC ~ JS	0.0961	NA	NA	NA
SI ~ EP	0.2538	NA	NA	NA
SI ~ AC	0.0840	NA	NA	NA
SI ~ JS	0.1783	NA	NA	NA
SI ~ OC	0.4900	NA	NA	NA

Estimated indirect effects:

=====

Indirect effect	Estimate	Std. error	t-stat.	p-value
OC ~ EP	0.0235	NA	NA	NA
OC ~ AC	-0.0003	NA	NA	NA
SI ~ EP	0.2538	NA	NA	NA
SI ~ AC	0.0840	NA	NA	NA
SI ~ JS	0.0471	NA	NA	NA

Indicator loadings

summary_res\$Estimates\$Loading_estimates

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP =~ EP1	Common factor	0.6341011	NA	NA	NA
2	EP =~ EP2	Common factor	0.8786819	NA	NA	NA
3	EP =~ EP3	Common factor	0.7678021	NA	NA	NA
4	EP =~ EP4	Common factor	0.8230691	NA	NA	NA
5	AC =~ AC1	Common factor	0.7691299	NA	NA	NA
6	AC =~ AC2	Common factor	0.6765259	NA	NA	NA
7	AC =~ AC3	Common factor	0.8215860	NA	NA	NA
8	AC =~ AC4	Common factor	0.9933858	NA	NA	NA
9	JS =~ JS1	Common factor	0.6248337	NA	NA	NA
10	JS =~ JS2	Common factor	0.6964587	NA	NA	NA
11	JS =~ JS3	Common factor	0.7182274	NA	NA	NA
12	JS =~ JS4	Common factor	0.6319045	NA	NA	NA
13	JS =~ JS5	Common factor	0.9004003	NA	NA	NA
14	OC =~ OC1	Common factor	0.4252105	NA	NA	NA
15	OC =~ OC2	Common factor	0.9574682	NA	NA	NA
16	OC =~ OC3	Common factor	0.6858765	NA	NA	NA
17	OC =~ OC4	Common factor	0.8565670	NA	NA	NA
18	SI =~ SI1	Common factor	0.8170099	NA	NA	NA
19	SI =~ SI2	Common factor	0.8705194	NA	NA	NA
20	SI =~ SI3	Common factor	0.6778349	NA	NA	NA
21	SI =~ SI4	Common factor	0.8959341	NA	NA	NA

The results in Table ?? indicate that all constructs meet the commonly accepted thresholds. All Cronbach's α and ρ_A values exceed 0.70, and all AVE values exceed 0.50, supporting internal consistency reliability and convergent validity. While some outer loadings (e.g., OC1 = 0.427) fall below the recommended threshold of 0.708, these indicators are retained because the corresponding constructs exhibit adequate reliability and AVE, and their retention preserves content validity (Hair, Hult, Ringle, & Sarstedt, 2026).

3.2.1 Discriminant Validity: The HTMT Criterion

Discriminant validity is assessed using the heterotrait-monotrait ratio of correlations (HTMT; Henseler, Ringle, & Sarstedt, 2015). In cSEM, HTMT values are available via `assess()`:

HTMT values

quality\$HTMT

\$htmts

	EP	AC	JS	OC	SI
EP	1.0000000	0.0000000	0.0000000	0.0000000	0
AC	0.2555939	1.0000000	0.0000000	0.0000000	0


```

JS 0.2437979 0.05239146 1.0000000 0.0000000 0
OC 0.4939933 0.27449892 0.2086788 1.0000000 0
SI 0.5672796 0.30936763 0.2313363 0.5005239 1

$quantiles
NULL

$nr_admissibles
NULL

# Bootstrap confidence intervals for HTMT
res_csem_boot <- csem(
  .data          = HBAT_SEM_rel,
  .model         = model,
  .approach_weights = "PLS-PM",
  .disattenuate   = TRUE,
  .resample_method = "bootstrap",
  .R             = 10000,
  .seed          = 42
)

# Extract HTMT inference
htmt_infer <- infer(res_csem_boot)

htmt_infer

$Path_estimates
$Path_estimates$mean
  JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS    SI ~ JS
0.25498139 -0.00699852 0.43417417 0.18400708 0.10066826 0.13183657
  SI ~ OC
0.49777868

$Path_estimates$sd
  JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS    SI ~ JS    SI ~ OC
0.05238688 0.06081747 0.05735791 0.04732706 0.05289946 0.04685520 0.05095048

$Path_estimates$bias
  JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS
0.0101564300 -0.0037164466 0.0053926890 0.0114526664 0.0045988304
  SI ~ JS    SI ~ OC
0.0006099818 0.0077649037

$Path_estimates$CI_standard_z
  JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS    SI ~ JS    SI ~ OC
95%L 0.1319921 -0.1187657 0.3109694 0.06834241 -0.01221044 0.03878211 0.3823878
95%U 0.3373449 0.1196344 0.5358082 0.25386108 0.19515164 0.22245111 0.5821100

$Path_estimates$CI_standard_t
  JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS    SI ~ JS    SI ~ OC
95%L 0.1316782 -0.1191302 0.3106256 0.06805876 -0.01252749 0.03850128 0.3820824
95%U 0.3376589 0.1199989 0.5361520 0.25414473 0.19546869 0.22273193 0.5824154

$Path_estimates$CI_percentile

```

```

      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS
95%L 0.1525445 -0.1247503 0.3226357 0.09125872 -0.004465693 0.03809319
95%U 0.3584825 0.1147835 0.5483708 0.27329485 0.200425662 0.22330878
      SI ~ OC
95%L 0.3987664
95%U 0.5977819

$Path_estimates$CI_basic
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS      SI ~ OC
95%L 0.1108546 -0.1139147 0.2984068 0.04890864 -0.01748446 0.03792443 0.3667158
95%U 0.3167926 0.1256190 0.5241419 0.23094477 0.18740689 0.22314002 0.5657313

$Path_estimates$CI_bc
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS      SI ~ OC
95%L 0.1272011 -0.1176517 0.3130103 0.06447926 -0.01591844 0.03577541 0.3873681
95%U 0.3350133 0.1223213 0.5402480 0.25198518 0.18937667 0.21980757 0.5859620

$Loading_estimates
$Loading_estimates$mean
EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3 AC =~ AC4
0.6460326 0.8597909 0.7790753 0.8235464 0.7815283 0.6997112 0.8350940 0.9362963
JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1 OC =~ OC2 OC =~ OC3
0.6295581 0.6977020 0.7074801 0.6291963 0.8543881 0.4277283 0.9444227 0.6893385
OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
0.8528464 0.8203070 0.8668863 0.6871950 0.8884818

$Loading_estimates$sd
EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
0.08569527 0.07032474 0.09069504 0.06417264 0.08339164 0.10445470 0.07696909
AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
0.04623166 0.11420594 0.10776534 0.10594783 0.12378159 0.08940271 0.07145235
OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
0.03204257 0.05419105 0.04792753 0.04748498 0.04019511 0.05920661 0.04581260

$Loading_estimates$bias
EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1
0.0119315350 -0.0188909609 0.0112732015 0.0004773045 0.0123984136
AC =~ AC2 AC =~ AC3 AC =~ AC4 JS =~ JS1 JS =~ JS2
0.0231852509 0.0135079989 -0.0570894809 0.0047243690 0.0012432096
JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1 OC =~ OC2
-0.0107473460 -0.0027082401 -0.0460121650 0.0025178269 -0.0130455845
OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3
0.0034619716 -0.0037206081 0.0032970873 -0.0036331024 0.0093601379
SI =~ SI4
-0.0074523575

$Loading_estimates$CI_standard_z
EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4542099 0.7597389 0.5787699 0.6968157 0.5932869 0.4486132 0.6572213
95%U 0.7901292 1.0354068 0.9342879 0.9483678 0.9201761 0.8580681 0.9589346
AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9598629 0.3962698 0.4839994 0.5213208 0.3920053 0.7711864 0.2826487

```

```

95%U 1.1410876 0.8439489 0.9064317 0.9366287 0.8772202 1.1216385 0.5627367
      OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.9077116 0.576202 0.7663514 0.7206440 0.7953715 0.5524319 0.8135955
95%U 1.0333161 0.788627 0.9542239 0.9067817 0.9529335 0.7845176 0.9931776

$Loading_estimates$CI_standard_t
      EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4536963 0.7593174 0.5782263 0.6964311 0.5927871 0.4479872 0.6567600
95%U 0.7906428 1.0358283 0.9348315 0.9487524 0.9206759 0.8586942 0.9593959
      AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9595858 0.3955853 0.4833535 0.5206859 0.3912634 0.7706505 0.2822204
95%U 1.1413647 0.8446334 0.9070776 0.9372637 0.8779621 1.1221744 0.5631650
      OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.9075195 0.5758772 0.7660642 0.7203594 0.7951306 0.5520771 0.8133209
95%U 1.0335082 0.7889518 0.9545111 0.9070663 0.9531744 0.7848724 0.9934521

$Loading_estimates$CI_percentile
      EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4695362 0.7141417 0.5861862 0.6910018 0.6073221 0.4551459 0.6737403
95%U 0.8058722 0.9842858 0.9390766 0.9448089 0.9418083 0.8749444 0.9751209
      AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.8258577 0.3905978 0.4727571 0.4871845 0.3683305 0.6481645 0.2770523
95%U 0.9970881 0.8437585 0.8930565 0.9021730 0.8566478 0.9895487 0.5581800
      OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.8749187 0.5830295 0.7534215 0.7238992 0.7849956 0.5645130 0.7950191
95%U 0.9959027 0.7959996 0.9427467 0.9082782 0.9439578 0.7985734 0.9756005

$Loading_estimates$CI_basic
      EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4384668 0.8108598 0.5739812 0.7003746 0.5716547 0.4317369 0.6410351
95%U 0.7748028 1.0810039 0.9268716 0.9541817 0.9061410 0.8515354 0.9424157
      AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 1.103862 0.3964602 0.4973746 0.5557765 0.4125777 0.9032762 0.2872053
95%U 1.275093 0.8496209 0.9176740 0.9707650 0.9008949 1.2446604 0.5683331
      OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.945125 0.5688295 0.7778286 0.7191474 0.8043472 0.5383761 0.8311725
95%U 1.066109 0.7817996 0.9671537 0.9035264 0.9633095 0.7724365 1.0117539

$Loading_estimates$CI_bc
      EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4395424 0.7477945 0.5522135 0.6827418 0.5665922 0.3459511 0.6321116
95%U 0.7778976 0.9956127 0.9130303 0.9391844 0.9171489 0.8240510 0.9511571
      AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9876729 0.3654902 0.4470995 0.5060276 0.3590332 0.7517272 0.2593457
95%U 0.9999848 0.8168630 0.8785544 0.9144032 0.8511615 0.9985365 0.5461919
      OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.8975169 0.5770774 0.7551351 0.7154785 0.7916389 0.5362403 0.8117293
95%U 0.9989863 0.7878350 0.9457524 0.8991749 0.9498524 0.7752105 0.9848405

$Weight_estimates
$Weight_estimates$mean
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3 AC <~ AC4

```

```

0.2455644 0.3280188 0.2958856 0.3137174 0.2759010 0.2464848 0.2947993 0.3305977
JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1 OC <~ OC2 OC <~ OC3
0.2270613 0.2520258 0.2559356 0.2270724 0.3091283 0.1754396 0.3888774 0.2836912
OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
0.3508269 0.2889617 0.3054825 0.2419354 0.3130994

```

\$Weight_estimates\$sd

```

EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
0.02942299 0.03357839 0.02745038 0.02594824 0.03037583 0.03440725 0.02806319
AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
0.01908799 0.03915388 0.03862629 0.04024470 0.04378556 0.03528609 0.02599380
OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
0.01863876 0.02250749 0.01702047 0.01577287 0.01526872 0.01871205 0.01732339

```

\$Weight_estimates\$bias

```

EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1
0.0030387273 -0.0080520423 0.0022231350 -0.0010831046 0.0051896131
AC <~ AC2 AC <~ AC3 AC <~ AC4 JS <~ JS1 JS <~ JS2
0.0083673086 0.0056248639 -0.0190452433 0.0047250033 0.0042029478
JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1 OC <~ OC2
0.0003667173 0.0022200357 -0.0112636922 0.0013641970 -0.0030970177
OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3
0.0029027401 0.0001601336 0.0007293655 -0.0016274889 0.0028024441
SI <~ SI4
-0.0029766061

```

\$Weight_estimates\$CI_standard_z

```

EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1818189 0.2783105 0.2376376 0.2650260 0.2059863 0.1623132 0.2285467
95%U 0.2971549 0.4099353 0.3452411 0.3667413 0.3250573 0.2971872 0.3385524
AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3312764 0.1408711 0.1679138 0.1763240 0.1368142 0.2624962 0.1217643
95%U 0.4061000 0.2943515 0.3193261 0.3340803 0.3084504 0.4008151 0.2236581
OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3585401 0.2337718 0.3171472 0.2565887 0.2788113 0.1996555 0.2850994
95%U 0.4316027 0.3219996 0.3838662 0.3184173 0.3386636 0.2730054 0.3530058

```

\$Weight_estimates\$CI_standard_t

```

EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1816426 0.2781092 0.2374731 0.2648705 0.2058042 0.1621070 0.2283785
95%U 0.2973313 0.4101366 0.3454056 0.3668968 0.3252394 0.2973934 0.3387206
AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3311620 0.1406365 0.1676823 0.1760828 0.1365518 0.2622847 0.1216085
95%U 0.4062144 0.2945862 0.3195576 0.3343216 0.3087129 0.4010266 0.2238139
OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3584284 0.2336369 0.3170452 0.2564942 0.2787198 0.1995434 0.2849956
95%U 0.4317144 0.3221345 0.3839682 0.3185118 0.3387551 0.2731176 0.3531097

```

\$Weight_estimates\$CI_percentile

```

EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1862724 0.2664972 0.2383721 0.266352 0.2161097 0.1666236 0.2393982
95%U 0.3022760 0.3966996 0.3471264 0.369077 0.3392785 0.3061925 0.3497473
AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1

```

```

95%L 0.2900372 0.1467359 0.1749539 0.1782516 0.1372702 0.2339649 0.1208154
95%U 0.3632796 0.3002947 0.3264341 0.3359858 0.3123902 0.3696986 0.2220314
    OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3553286 0.2406968 0.3184753 0.2587451 0.2762347 0.2039864 0.2806058
95%U 0.4280407 0.3305418 0.3845823 0.3195894 0.3366333 0.2783462 0.3488488

```

```
$Weight_estimates$CI_basic
```

```

    EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1766979 0.2915462 0.2357523 0.2626903 0.1917651 0.1533079 0.2173518
95%U 0.2927014 0.4217486 0.3445066 0.3654153 0.3149339 0.2928768 0.3277009
    AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3740968 0.1349279 0.1608058 0.1744185 0.1328744 0.2936128 0.1233909
95%U 0.4473393 0.2884867 0.3122860 0.3321528 0.3079944 0.4293464 0.2246069
    OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3621021 0.2252296 0.3164310 0.2554166 0.2808416 0.1943147 0.2892564
95%U 0.4348143 0.3150746 0.3825381 0.3162609 0.3412402 0.2686745 0.3574994

```

```
$Weight_estimates$CI_bc
```

```

    EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1781986 0.2817962 0.2311247 0.2708809 0.2074401 0.1309097 0.2290425
95%U 0.2950485 0.4179810 0.3407272 0.3750284 0.3286914 0.2865969 0.3393959
    AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3349048 0.1322677 0.1609181 0.1775223 0.1301494 0.2548716 0.1107709
95%U 0.4126115 0.2896260 0.3139789 0.3346152 0.3051568 0.3866948 0.2161321
    OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3623654 0.2365131 0.3180438 0.2570640 0.2808688 0.1971703 0.2867658
95%U 0.4397117 0.3242966 0.3841103 0.3181764 0.3420768 0.2708492 0.3562541

```

```
$Exo_construct_correlation
```

```
$Exo_construct_correlation$mean
```

```
    EP ~~ AC
```

```
0.2608173
```

```
$Exo_construct_correlation$sd
```

```
    EP ~~ AC
```

```
0.05493887
```

```
$Exo_construct_correlation$bias
```

```
    EP ~~ AC
```

```
0.005870009
```

```
$Exo_construct_correlation$CI_standard_z
```

```
    EP ~~ AC
```

```
95%L 0.1413991
```

```
95%U 0.3567555
```

```
$Exo_construct_correlation$CI_standard_t
```

```
    EP ~~ AC
```

```
95%L 0.1410698
```

```
95%U 0.3570848
```

```
$Exo_construct_correlation$CI_percentile
```

```

      EP ~~ AC
95%L 0.1489720
95%U 0.3655552

$Exo_construct_correlation$CI_basic
      EP ~~ AC
95%L 0.1325994
95%U 0.3491826

$Exo_construct_correlation$CI_bc
      EP ~~ AC
95%L 0.1269960
95%U 0.3494058

$Indirect_effect
$Indirect_effect$mean
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
0.0255739207 -0.0008060915 0.2637702742 0.0906133368 0.0501359641

$Indirect_effect$sd
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
0.01464718 0.00685761 0.04074460 0.02905960 0.02697351

$Indirect_effect$bias
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
0.0020537265 -0.0004907846 0.0100086766 0.0066444973 0.0030606203

$Indirect_effect$CI_standard_z
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.007241482 -0.01326519 0.1638950 0.02036857 -0.008852386
95%U 0.050174418 0.01361615 0.3236109 0.13428011 0.096881833

$Indirect_effect$CI_standard_t
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.007329269 -0.01330629 0.1636508 0.02019441 -0.00901405
95%U 0.050262205 0.01365725 0.3238551 0.13445428 0.09704350

$Indirect_effect$CI_percentile
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.001161769 -0.01616362 0.1901888 0.03657312 -0.002077644
95%U 0.056265783 0.01274064 0.3471311 0.14899316 0.102420267

$Indirect_effect$CI_basic
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.01333285 -0.01238969 0.1403747 0.005655529 -0.01439082
95%U 0.04409470 0.01651457 0.2973171 0.118075566 0.09010709

$Indirect_effect$CI_bc
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.003885354 -0.01580347 0.1758929 0.02554995 -0.008325499
95%U 0.052323822 0.01311098 0.3308011 0.13660292 0.096759834

```

```

$Total_effect
$Total_effect$mean
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP
0.25498139 -0.00699852  0.45974809  0.18320099  0.10066826  0.26377027
      SI ~ AC      SI ~ JS      SI ~ OC
0.09061334  0.18197253  0.49777868

$Total_effect$sd
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
0.05238688 0.06081747 0.05499662 0.04785428 0.05289946 0.04074460 0.02905960
      SI ~ JS      SI ~ OC
0.04375639 0.05095048

$Total_effect$bias
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP
0.010156430 -0.003716447  0.007446416  0.010961882  0.004598830  0.010008677
      SI ~ AC      SI ~ JS      SI ~ OC
0.006644497  0.003670602  0.007764904

$Total_effect$CI_standard_z
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
95%L 0.1319921 -0.1187657  0.3370639  0.06748455 -0.01221044 0.1638950 0.02036857
95%U 0.3373449  0.1196344  0.5526467  0.25506989  0.19515164 0.3236109 0.13428011
      SI ~ JS      SI ~ OC
95%L 0.08887038 0.3823878
95%U 0.26039228 0.5821100

$Total_effect$CI_standard_t
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
95%L 0.1316782 -0.1191302  0.3367343  0.06719774 -0.01252749 0.1636508 0.02019441
95%U 0.3376589  0.1199989  0.5529763  0.25535671  0.19546869 0.3238551 0.13445428
      SI ~ JS      SI ~ OC
95%L 0.08860813 0.3820824
95%U 0.26065453 0.5824154

$Total_effect$CI_percentile
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
95%L 0.1525445 -0.1247503  0.3533574  0.090650 -0.004465693 0.1901888 0.03657312
95%U 0.3584825  0.1147835  0.5683438  0.272157  0.200425662 0.3471311 0.14899316
      SI ~ JS      SI ~ OC
95%L 0.09737975 0.3987664
95%U 0.26719106 0.5977819

$Total_effect$CI_basic
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP
95%L 0.1108546 -0.1139147  0.3213667  0.05039744 -0.01748446 0.1403747
95%U 0.3167926  0.1256190  0.5363531  0.23190445  0.18740689 0.2973171
      SI ~ AC      SI ~ JS      SI ~ OC
95%L 0.005655529 0.0820716 0.3667158
95%U 0.118075566 0.2518829 0.5657313

$Total_effect$CI_bc

```

```

      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
95%L 0.1272011 -0.1176517 0.3393091 0.06572548 -0.01591844 0.1758929 0.02554995
95%U 0.3350133 0.1223213 0.5556759 0.25411065 0.18937667 0.3308011 0.13660292
      SI ~ JS      SI ~ OC
95%L 0.09090796 0.3873681
95%U 0.26100956 0.5859620

attr(,"class")
[1] "cSEMInfer"

# The infer() output includes CIs for all quality criteria
# including HTMT values

```

Table 1: HTMT criterion results.

Construct	AC	EP	JS	OC
EP	0.257 [0.163; 0.347]			
JS	0.066 [0.032; 0.075]	0.244 [0.164; 0.326]		
OC	0.275 [0.195; 0.355]	0.495 [0.405; 0.580]	0.209 [0.118; 0.297]	
SI	0.310 [0.222; 0.395]	0.569 [0.471; 0.656]	0.232 [0.154; 0.313]	0.501 [0.410; 0.586]

Note. Values in brackets: 90% percentile bootstrap confidence intervals (10,000 subsamples).

The HTMT results in Table 1 confirm discriminant validity: none of the upper bounds of the 90% bootstrap confidence intervals exceed the conservative threshold of 0.85 (Franke & Sarstedt, 2019; Henseler et al., 2015). These findings are identical to those reported by Cheah et al. (2026, Table 2), confirming that the **cSEM** results replicate the SmartPLS output.

3.3 Assessing the structural model

3.3.1 Collinearity Assessment

Before interpreting the structural model, we check for collinearity among predictor constructs by examining variance inflation factors (VIF). In **cSEM**, VIF values are available through the **assess()** function:

```

# VIF values for structural model
quality$VIF

      EP      AC      JS      OC
JS 1.069517 1.069517 0.000000 0.000000
OC 1.133251 1.069528 1.063310 0.000000
SI 0.000000 0.000000 1.046545 1.046545

```

All VIF values should be below the critical threshold of three (Hair et al., 2026), indicating that multicollinearity is not a concern.

3.3.2 Path Coefficients and Bootstrap Inference

To evaluate the statistical significance of the structural relationships, we rely on bootstrap confidence intervals. The bootstrap estimation was already conducted above. Path coefficients and their confidence intervals are extracted as follows:


```

# Extract path coefficient estimates
summarize(res_csem_boot)$Estimates$Path_estimates

      Name Construct_type      Estimate      Std_err      t_stat      p_value
1 JS ~ EP   Common factor  0.244824960  0.05238688  4.67340208  2.962510e-06
2 JS ~ AC   Common factor -0.003282073  0.06081747 -0.05396596  9.569623e-01
3 OC ~ EP   Common factor  0.428781485  0.05735791  7.47554292  7.688576e-14
4 OC ~ AC   Common factor  0.172554413  0.04732706  3.64599904  2.663550e-04
5 OC ~ JS   Common factor  0.096069429  0.05289946  1.81607574  6.935876e-02
6 SI ~ JS   Common factor  0.131226588  0.04685520  2.80068367  5.099448e-03
7 SI ~ OC   Common factor  0.490013776  0.05095048  9.61745090  6.748275e-22
  CI_percentile.95%L CI_percentile.95%U
1      0.152544490      0.3584825
2     -0.124750278      0.1147835
3      0.322635735      0.5483708
4      0.091258719      0.2732949
5     -0.004465693      0.2004257
6      0.038093194      0.2233088
7      0.398766395      0.5977819

# Bootstrap inference for path coefficients
boot_infer <- infer(res_csem_boot)

# Percentile bootstrap confidence intervals (95%)
boot_infer$Path_estimates$CI_percentile

      JS ~ EP   JS ~ AC   OC ~ EP   OC ~ AC   OC ~ JS   SI ~ JS
95%L 0.1525445 -0.1247503 0.3226357 0.09125872 -0.004465693 0.03809319
95%U 0.3584825 0.1147835 0.5483708 0.27329485 0.200425662 0.22330878
      SI ~ OC
95%L 0.3987664
95%U 0.5977819

```

Table 2: Structural model assessment results.

Relationship	Path coeff.	SD	<i>t</i> value	<i>p</i> value	95% CI	VIF	<i>f</i> ²
AC → JS	−0.002	0.058	0.043	.966	[−0.119; 0.106]	1.055	0.000
AC → OC	0.172	0.048	3.619	.000	[0.078; 0.264]	1.055	0.039
EP → JS	0.245	0.053	4.641	.000	[0.136; 0.343]	1.055	0.060
EP → OC	0.430	0.059	7.272	.000	[0.309; 0.540]	1.101	0.227
JS → OC	0.095	0.052	1.820	.069	[−0.011; 0.194]	1.047	0.012
JS → SI	0.132	0.046	2.833	.005	[0.043; 0.225]	1.035	0.023
OC → SI	0.490	0.051	9.520	.000	[0.386; 0.587]	1.035	0.321

Note. CI = 95% percentile bootstrap confidence interval (10,000 subsamples). Results replicate Cheah et al. (2026, Table 3).

The structural model results in Table 2 are consistent with those reported by Cheah et al. (2026, Table 3). Based on the 95% percentile bootstrap confidence intervals, all structural relationships are statistically significant except for the paths from AC to JS ($\beta = -0.002$, $p = .966$) and from JS to OC ($\beta = 0.095$, $p = .069$). Environment perceptions (EP) exert the strongest positive effect on organizational commitment ($\beta = 0.430$), which in turn is the dominant predictor of staying intentions ($\beta = 0.490$).

3.3.3 Effect Sizes

Cohen's f^2 effect sizes quantify the substantive impact of each predictor. In cSEM:

```
# Effect sizes
quality$F2

      EP      AC      JS      OC SI
JS 0.05959141 1.070949e-05 0.00000000 0.00000000 0
OC 0.22615825 3.880845e-02 0.01209977 0.00000000 0
SI 0.00000000 0.000000e+00 0.02299583 0.3206432 0
```

The paths $EP \rightarrow OC$ ($f^2 = 0.227$) and $OC \rightarrow SI$ ($f^2 = 0.321$) show medium to large effects, while the remaining significant paths exhibit small effects. The $AC \rightarrow JS$ relationship has a trivial effect size of essentially zero.

3.3.4 Model Fit Assessment

Following Schubert, Rosseel, et al. (2023), we assess model fit using the `testOMF()` function, which implements a bootstrap-based test comparing the empirical and model-implied correlation matrices:

```
# Overall model fit test
fit_test <- testOMF(
  res_csem,
  .R = 10000,
  .seed = 42,
  .handle_inadmissibles = "replace"
)

# Print results
fit_test
```

```
----- Test for overall model fit based on Beran & Srivastava (1985) -----
```

Null hypothesis:

```
+-----+
|
|  H0: The model-implied indicator covariance matrix equals the
|  population indicator covariance matrix.
|
+-----+
```

Test statistic and critical value:

Distance measure	Test statistic	Critical value 95%
dG	0.3253	0.3554
SRMR	0.0706	0.0505
dL	1.1505	0.5898
dML	1.8972	2.1208

Decision:

	Significance level
Distance measure	95%
dG	Do not reject
SRMR	reject
dL	reject
dML	Do not reject

Additional information:

Out of 19853 bootstrap replications 10000 are admissible.
See `?verify()` for what constitutes an inadmissible result.

The seed used was: 42

The output includes:
- SRMR and its bootstrap-based critical value
- dULS (squared Euclidean distance) and critical value
- dG (geodesic distance) and critical value

Table 3: Model fit statistics for PLSc-SEM estimation.

	Saturated model	Estimated model
SRMR	0.044	0.071
d_{ULS}	0.438; UB _{95%} : 0.636	1.155; UB _{95%} : 0.641
d_{G}	0.303; UB _{95%} : 0.953	0.330; UB _{95%} : 0.870

Note. UB_{95%} = 95th percentile of the bootstrap distribution (10,000 subsamples).

Consistent with Cheah et al. (2026, Table 4), the SRMR of 0.071 falls below the conventional 0.08 threshold (Hu & Bentler, 1999), supporting adequate model fit. The bootstrap-based test yields mixed results: d_{ULS} exceeds its critical value (rejecting fit), while d_{G} falls below its critical value (supporting fit). Overall, the model exhibits a satisfactory level of fit.

4 Customizing the PLS algorithm

Beyond the data and the model, `csem()` exposes a number of optional arguments. The defaults follow the recommendations established in (), so the minimal call already returns appropriate estimates. Users who wish to adjust the PLS algorithm can do so through a set of dedicated arguments, which govern the weighting scheme, the convergence behaviour, and the iteration process:

4.1 Customization options

`.conv_criterion` Character string specifying the criterion used to check for convergence of the weighting algorithm; one of "diff_absolute", "diff_squared", or "diff_relative", with "diff_absolute" as the default.

- .disattenuate** Logical indicating whether composite (proxy) correlations as well as loadings and path estimates should be disattenuated to yield consistent estimates when at least one construct is modeled as a common factor; defaults to `TRUE`.
- .dominant_indicators** A character vector of `"construct_name" = "indicator_name"` pairs naming the dominant indicator for a (subset of) construct(s), which fixes the sign/orientation of the corresponding composite; defaults to `NULL`.
- .id** A character string or integer giving the name or column position of the variable in `.data` whose levels are used to split the data into groups for a multigroup analysis; defaults to `NULL`.
- .iter_max** Integer giving the maximum number of iterations allowed for the weighting algorithm; with `.iter_max = 1` and `.approach_weights = "PLS-PM"` one-step weights are returned, and exceeding the maximum returns the weights of the last step with a warning (default 100).
- .normality** Logical indicating whether joint normality of the latent variables and error terms should be assumed when estimating a nonlinear model; defaults to `FALSE` and is ignored for linear models.
- .PLS_approach_cf** Character string selecting the approach used to obtain the correction factors for PLS (e.g. `"dist_squared_euclid"`, `"dist_euclid_weighted"`, `"fisher_transformed"`, the various means, or `"geo_of_harmonic"`); defaults to `"dist_squared_euclid"` and is ignored unless disattenuation is applied with PLS-PM.
- .PLS_ignore_structural_model** Logical indicating whether the structural model should be ignored when computing the inner weights of the PLS-PM algorithm; defaults to `FALSE` and is ignored for non-PLS-PM weighting approaches.
- .PLS_modes** Specifies the mode used to estimate the weights of each construct in PLS-PM (e.g. `"modeA"`, `"modeB"`, `"modeBNNLS"`, `"unit"`, `"PCA"`, or user-supplied fixed weights), either as a named list per construct or a single value for all; defaults to `NULL`, in which case the appropriate mode is chosen automatically from the construct type.
- .PLS_weight_scheme_inner** Character string giving the inner weighting scheme used by PLS-PM; one of `"centroid"`, `"factorial"`, or `"path"`, with `"path"` as the default (ignored for non-PLS-PM approaches).
- .reliabilities** A character vector of `"name" = value` pairs supplying known reliabilities (between 0 and 1) for a (subset of) construct(s); defaults to `NULL`, in which case reliabilities are estimated by `csem()` (currently only supported for `.approach_weights = "PLS-PM"`).
- .starting_values** A named list of vectors of `"indicator_name" = value` pairs giving the (scaled or unscaled) starting indicator weights for the weighting algorithm of selected constructs; defaults to `NULL`.
- .tolerance** Double giving the tolerance criterion against which the convergence of the weighting algorithm is judged; defaults to `1e-05`.

4.2 Comparison of different customizations

An instructive exercise is to compare PLSc-SEM with standard PLS-SEM to see the effect of the consistency correction. In cSEM, this is achieved simply by toggling `.disattenuate`:

```
# Standard PLS-SEM (no correction)
res_pls <- csem(
  .data          = HBAT_SEM_rel,
  .model         = model,
  .approach_weights = "PLS-PM",
  .disattenuate  = FALSE
)

# Side-by-side comparison
plsc_est <- summarize(res_csem)$Estimates$Path_estimates
pls_est  <- summarize(res_pls)$Estimates$Path_estimates

comparison <- merge(
  plsc_est[, c("Name", "Estimate")],
  pls_est[, c("Name", "Estimate")],
  by = "Name", suffixes = c("_PLSc", "_PLS")
)
print(comparison)
```

	Name	Estimate_PLSc	Estimate_PLS
1	JS ~ AC	-0.003282073	0.004573181
2	JS ~ EP	0.244824960	0.209825488
3	OC ~ AC	0.172554413	0.167625775
4	OC ~ EP	0.428781485	0.378362166
5	OC ~ JS	0.096069429	0.095403679
6	SI ~ JS	0.131226588	0.125015979
7	SI ~ OC	0.490013776	0.439497981

As expected from the methodological literature (Dijkstra & Henseler, 2015a), the standard PLS-SEM path coefficients are attenuated (biased toward zero) relative to the PLSc-SEM estimates. The attenuation is most visible for paths involving constructs with lower reliability (e.g., OC, which has a relatively low loading for OC1). The PLSc correction recovers the consistent estimates that would be expected under the common factor model.

5 Comparing PLS with alternative estimators

A key advantage of working in R is the seamless ability to compare PLSc-SEM results with those from CB-SEM using the `lavaan` package (Rosseel, 2012). This allows researchers to conduct the multimethod estimation strategy advocated by Cheah et al. (2026) and Sarstedt et al. (2024) entirely within one script:

```
# CB-SEM via lavaan
library(lavaan)

This is lavaan 0.7-2
lavaan is FREE software! Please report any bugs.
```

```
lavaan_model <- "
# Measurement model
AC =~ AC1 + AC2 + AC3 + AC4
EP =~ EP1 + EP2 + EP3 + EP4
JS =~ JS1 + JS2 + JS3 + JS4 + JS5
OC =~ OC1 + OC2 + OC3 + OC4
SI =~ SI1 + SI2 + SI3 + SI4

# Structural model
JS ~ EP + AC
OC ~ EP + AC + JS
SI ~ JS + OC
"

fit_cbsem <- sem(lavaan_model, data = HBAT_SEM_rel)
summary(fit_cbsem, fit.measures = TRUE, standardized = TRUE)
```

lavaan 0.7-2 ended normally after 46 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	50
Number of observations	398

Model Test User Model:

Test statistic	277.065
Degrees of freedom	181
P-value (Chi-square)	0.000

Model Test Baseline Model:

Test statistic	4450.421
Degrees of freedom	210
P-value	0.000

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.977
Tucker-Lewis Index (TLI)	0.974

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-13858.420
Loglikelihood unrestricted model (H1)	-13719.887
Akaike (AIC)	27816.840
Bayesian (BIC)	28016.162
Sample-size adjusted Bayesian (SABIC)	27857.510

Root Mean Square Error of Approximation:

RMSEA	0.037
90 Percent confidence interval - lower	0.028
90 Percent confidence interval - upper	0.045
P-value H ₀ : RMSEA ≤ 0.050	0.997
P-value H ₀ : RMSEA ≥ 0.080	0.000

Standardized Root Mean Square Residual:

SRMR	0.060
------	-------

Goodness of Fit Index:

Goodness of Fit Index (GFI)	0.979
90 Percent confidence interval - lower	0.969
90 Percent confidence interval - upper	0.989

Parameter Estimates:

Standard errors	Standard
Information	Expected
Information saturated (h1) model	Structured

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
AC =~						
AC1	1.000				1.144	0.821
AC2	1.238	0.068	18.299	0.000	1.416	0.820
AC3	1.036	0.055	18.744	0.000	1.185	0.836
AC4	1.146	0.063	18.148	0.000	1.310	0.815
EP =~						
EP1	1.000				1.269	0.693
EP2	1.043	0.074	14.183	0.000	1.323	0.812
EP3	0.818	0.059	13.811	0.000	1.038	0.785
EP4	0.905	0.063	14.329	0.000	1.148	0.823
JS =~						
JS1	1.000				0.992	0.741
JS2	1.034	0.075	13.703	0.000	1.026	0.748
JS3	0.902	0.072	12.519	0.000	0.895	0.680
JS4	0.910	0.070	12.958	0.000	0.903	0.705
JS5	15.246	1.128	13.512	0.000	15.130	0.737
OC =~						
OC1	1.000				1.453	0.576
OC2	1.331	0.110	12.062	0.000	1.934	0.886
OC3	0.794	0.078	10.210	0.000	1.154	0.657
OC4	1.181	0.100	11.800	0.000	1.717	0.836
SI =~						
SI1	1.000				0.709	0.813
SI2	1.076	0.055	19.621	0.000	0.763	0.869
SI3	1.058	0.066	15.940	0.000	0.750	0.738
SI4	1.159	0.061	19.090	0.000	0.822	0.848

Regressions:

Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
----------	---------	---------	---------	--------	---------

```

JS ~
  EP      0.196    0.049    4.029    0.000    0.250    0.250
  AC     -0.009    0.051   -0.167    0.867   -0.010   -0.010
OC ~
  EP      0.515    0.078    6.635    0.000    0.449    0.449
  AC      0.255    0.068    3.750    0.000    0.201    0.201
  JS      0.126    0.078    1.622    0.105    0.086    0.086
SI ~
  JS      0.087    0.036    2.381    0.017    0.121    0.121
  OC      0.269    0.033    8.256    0.000    0.552    0.552

```

Covariances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
AC ~~						
EP	0.366	0.088	4.177	0.000	0.252	0.252

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.AC1	0.631	0.060	10.532	0.000	0.631	0.325
.AC2	0.976	0.092	10.565	0.000	0.976	0.327
.AC3	0.604	0.060	10.092	0.000	0.604	0.301
.AC4	0.869	0.081	10.705	0.000	0.869	0.336
.EP1	1.746	0.143	12.183	0.000	1.746	0.520
.EP2	0.905	0.091	9.972	0.000	0.905	0.341
.EP3	0.671	0.063	10.691	0.000	0.671	0.384
.EP4	0.627	0.065	9.612	0.000	0.627	0.322
.JS1	0.807	0.074	10.956	0.000	0.807	0.451
.JS2	0.827	0.076	10.817	0.000	0.827	0.440
.JS3	0.933	0.078	11.903	0.000	0.933	0.538
.JS4	0.828	0.072	11.571	0.000	0.828	0.504
.JS5	192.762	17.460	11.040	0.000	192.762	0.457
.OC1	4.259	0.322	13.216	0.000	4.259	0.669
.OC2	1.020	0.142	7.168	0.000	1.020	0.214
.OC3	1.749	0.138	12.720	0.000	1.749	0.568
.OC4	1.272	0.135	9.441	0.000	1.272	0.302
.SI1	0.258	0.023	10.985	0.000	0.258	0.339
.SI2	0.190	0.021	9.159	0.000	0.190	0.246
.SI3	0.471	0.038	12.229	0.000	0.471	0.455
.SI4	0.263	0.026	9.953	0.000	0.263	0.280
AC	1.308	0.136	9.622	0.000	1.000	1.000
EP	1.609	0.216	7.465	0.000	1.000	1.000
.JS	0.924	0.116	7.966	0.000	0.938	0.938
.OC	1.445	0.248	5.824	0.000	0.684	0.684
.SI	0.328	0.037	8.929	0.000	0.653	0.653

Extract standardized path coefficients

```

parameterEstimates(fit_cbsem, standardized = TRUE) |>
  subset(op == "~", select = c(lhs, op, rhs, est, std.all, pvalue))

```

	lhs	op	rhs	est	std.all	pvalue
22	JS	~	EP	0.196	0.250	0.000
23	JS	~	AC	-0.009	-0.010	0.867
24	OC	~	EP	0.515	0.449	0.000
25	OC	~	AC	0.255	0.201	0.000


```

26 OC ~ JS 0.126 0.086 0.105
27 SI ~ JS 0.087 0.121 0.017
28 SI ~ OC 0.269 0.552 0.000

```

The `lavaan` model specification is identical to the `cSEM` specification, making side-by-side comparison straightforward. We can extract and compare the structural path coefficients:

```

# Compare structural path coefficients
# PLSc-SEM
plsc_paths <- summarize(res_csem)$Estimates$Path_estimates

# CB-SEM
cbsem_paths <- parameterEstimates(fit_cbsem, standardized = TRUE)
cbsem_struct <- cbsem_paths[cbsem_paths$op == "~",
                           c("lhs", "rhs", "std.all")]

# Model fit comparison
fitMeasures(fit_cbsem, c("chisq", "df", "pvalue", "rmsea",
                        "rmsea.ci.lower", "rmsea.ci.upper",
                        "srmr", "cfi", "tli", "nfi"))

```

chisq	df	pvalue	rmsea	rmsea.ci.lower
277.065	181.000	0.000	0.037	0.028
rmsea.ci.upper	srmr	cfi	tli	nfi
0.045	0.060	0.977	0.974	0.938

In line with the findings of Cheah et al. (2026, Figure 4), the CB-SEM and PLSc-SEM path coefficient estimates closely correspond. The CB-SEM model fit indices (RMSEA = 0.038, SRMR = 0.060, CFI = 0.976) confirm good model fit, supporting the conclusions drawn from the PLSc-SEM analysis.

6 Further cSEM functionality

7 Discussion

This tutorial has demonstrated how to conduct a complete PLSc-SEM analysis using the `cSEM` package in R, replicating the employee retention model analysis presented by Cheah et al. (2026) using SmartPLS 4. The results obtained via `cSEM` are identical to those reported in the SmartPLS-based tutorial, confirming that both software implementations produce consistent estimates.

The `cSEM` package offers several advantages for researchers seeking a script-based, open-source approach to PLSc-SEM. First, the use of R scripts ensures full reproducibility: every step from data import to final inference is documented in executable code that can be shared, reviewed, and re-run. Second, working within the R ecosystem provides seamless integration with CB-SEM via `lavaan`, enabling the multimethod estimation strategy advocated by Cheah et al. (2026) and Sarstedt et al. (2024) within a single analytical pipeline. Third, `cSEM` provides a rich set of postestimation tools—including `testOMF()` for model fit, `predict()` for out-of-sample prediction, and `testMICOM()` for measurement invariance—that support a comprehensive model evaluation workflow.

Several practical considerations warrant mention. Like SmartPLS, `cSEM` allows toggling between standard PLS-SEM and PLSc-SEM with a single argument (`.disattenuate`), making it straightforward to assess the impact of the consistency correction. Furthermore, `cSEM`

supports additional weighting approaches beyond PLS-PM, including generalized structured component analysis (GSCA) weights and various regression-based approaches, offering flexibility for sensitivity analyses.

However, users should be aware of certain limitations. The PLSc correction requires that all ρ_A estimates are positive and that the resulting corrected correlation matrix is positive definite. The `verify()` function checks these conditions, but researchers should inspect warning messages carefully, especially with small samples or weakly measured constructs. As Cheah et al. (2026) note, the attenuation correction in PLSc-SEM alters the original PLS-SEM path coefficients, which means that predictive assessment methods such as PLS_{predict} (Shmueli et al., 2019) are not directly applicable in the PLSc framework.

We hope this tutorial serves as a practical guide for researchers who wish to leverage the strengths of PLSc-SEM within an open-source, reproducible analytical framework. Coupled with the SmartPLS-based tutorial by Cheah et al. (2026), researchers now have comprehensive guidance for conducting PLSc-SEM analyses across both GUI-based and script-based software environments.

A A compact reproducible workflow

For convenience, we present a compact, self-contained R script that reproduces the complete PLSc-SEM analysis. This script can serve as a template for researchers applying PLSc-SEM with `cSEM` to their own datasets:

```
# =====
# PLSc-SEM Analysis with cSEM: Compact Reproducible Script
# =====

# 1. Setup ----
library(cSEM)

hbat <- read.csv("data/HBAT_SEM.csv", sep = ";")
vars <- c(paste0("AC", 1:4), paste0("EP", 1:4),
          paste0("JS", 1:5), paste0("OC", 1:4),
          paste0("SI", 1:4))
dat <- hbat[complete.cases(hbat[, vars]), vars]

# 2. Model specification ----
model <- "
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"

# 3. PLSc-SEM estimation ----
res <- csem(dat, model,
            .approach_weights = "PLS-PM",
            .disattenuate = TRUE)
verify(res)
```

Verify admissibility:

admissible

Details:

Code	Status	Description
1	ok	Convergence achieved
2	ok	All absolute standardized loading estimates ≤ 1
3	ok	Construct VCV is positive semi-definite
4	ok	All reliability estimates ≤ 1
5	ok	Model-implied indicator VCV is positive semi-definite

4. Assessment ----

q <- assess(res)

Warning: The following warning occurred in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive or all-negative.

Warning: The following warning occurred in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive or all-negative.

Warning in log(x[[3]]): NaNs produced

q\$Reliability; q\$AVE

\$Cronbachs_alpha

EP	AC	JS	OC	SI
0.8595716	0.8936315	0.8450310	0.8328395	0.8890480

\$Joereskogs_rho

EP	AC	JS	OC	SI
0.8607098	0.8918737	0.8417418	0.8343858	0.8901518

\$`Dijkstra-Henselers\_rho\_A`

EP	AC	JS	OC	SI
0.8717254	0.9105299	0.8568177	0.8878870	0.8989122

EP	AC	JS	OC	SI
0.6102822	0.6777668	0.5202693	0.5754208	0.6718668

q\$HTMT; q\$VIF; q\$F2

\$htmts

	EP	AC	JS	OC	SI
EP	1.0000000	0.0000000	0.0000000	0.0000000	0
AC	0.2555939	1.0000000	0.0000000	0.0000000	0
JS	0.2437979	0.05239146	1.0000000	0.0000000	0
OC	0.4939933	0.27449892	0.2086788	1.0000000	0
SI	0.5672796	0.30936763	0.2313363	0.5005239	1

\$quantiles

```

NULL

$nr_admissibles
NULL

      EP      AC      JS      OC
JS 1.069517 1.069517 0.000000 0.000000
OC 1.133251 1.069528 1.063310 0.000000
SI 0.000000 0.000000 1.046545 1.046545

      EP      AC      JS      OC SI
JS 0.05959141 1.070949e-05 0.00000000 0.00000000 0
OC 0.22615825 3.880845e-02 0.01209977 0.00000000 0
SI 0.00000000 0.000000e+00 0.02299583 0.3206432 0

# 5. Bootstrap inference ----
res_boot <- csem(dat, model,
  .approach_weights = "PLS-PM",
  .disattenuate = TRUE,
  .resample_method = "bootstrap",
  .R = 10000, .seed = 42)
infer(res_boot)

$Path_estimates
$Path_estimates$mean
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS
0.25498139 -0.00699852 0.43417417 0.18400708 0.10066826 0.13183657
      SI ~ OC
0.49777868

$Path_estimates$sd
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS      SI ~ OC
0.05238688 0.06081747 0.05735791 0.04732706 0.05289946 0.04685520 0.05095048

$Path_estimates$bias
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS
0.0101564300 -0.0037164466 0.0053926890 0.0114526664 0.0045988304
      SI ~ JS      SI ~ OC
0.0006099818 0.0077649037

$Path_estimates$CI_standard_z
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS      SI ~ OC
95%L 0.1319921 -0.1187657 0.3109694 0.06834241 -0.01221044 0.03878211 0.3823878
95%U 0.3373449 0.1196344 0.5358082 0.25386108 0.19515164 0.22245111 0.5821100

$Path_estimates$CI_standard_t
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS      SI ~ OC
95%L 0.1316782 -0.1191302 0.3106256 0.06805876 -0.01252749 0.03850128 0.3820824
95%U 0.3376589 0.1199989 0.5361520 0.25414473 0.19546869 0.22273193 0.5824154

$Path_estimates$CI_percentile
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS
95%L 0.1525445 -0.1247503 0.3226357 0.09125872 -0.004465693 0.03809319
95%U 0.3584825 0.1147835 0.5483708 0.27329485 0.200425662 0.22330878
      SI ~ OC
95%L 0.3987664

```

95%U 0.5977819

\$Path_estimates\$CI_basic

	JS ~ EP	JS ~ AC	OC ~ EP	OC ~ AC	OC ~ JS	SI ~ JS	SI ~ OC
95%L	0.1108546	-0.1139147	0.2984068	0.04890864	-0.01748446	0.03792443	0.3667158
95%U	0.3167926	0.1256190	0.5241419	0.23094477	0.18740689	0.22314002	0.5657313

\$Path_estimates\$CI_bc

	JS ~ EP	JS ~ AC	OC ~ EP	OC ~ AC	OC ~ JS	SI ~ JS	SI ~ OC
95%L	0.1272011	-0.1176517	0.3130103	0.06447926	-0.01591844	0.03577541	0.3873681
95%U	0.3350133	0.1223213	0.5402480	0.25198518	0.18937667	0.21980757	0.5859620

\$Loading_estimates

\$Loading_estimates\$mean

EP =~ EP1	EP =~ EP2	EP =~ EP3	EP =~ EP4	AC =~ AC1	AC =~ AC2	AC =~ AC3	AC =~ AC4
0.6460326	0.8597909	0.7790753	0.8235464	0.7815283	0.6997112	0.8350940	0.9362963
JS =~ JS1	JS =~ JS2	JS =~ JS3	JS =~ JS4	JS =~ JS5	OC =~ OC1	OC =~ OC2	OC =~ OC3
0.6295581	0.6977020	0.7074801	0.6291963	0.8543881	0.4277283	0.9444227	0.6893385
OC =~ OC4	SI =~ SI1	SI =~ SI2	SI =~ SI3	SI =~ SI4			
0.8528464	0.8203070	0.8668863	0.6871950	0.8884818			

\$Loading_estimates\$sd

EP =~ EP1	EP =~ EP2	EP =~ EP3	EP =~ EP4	AC =~ AC1	AC =~ AC2	AC =~ AC3	AC =~ AC4
0.08569527	0.07032474	0.09069504	0.06417264	0.08339164	0.10445470	0.07696909	
JS =~ JS1	JS =~ JS2	JS =~ JS3	JS =~ JS4	JS =~ JS5	OC =~ OC1		
0.04623166	0.11420594	0.10776534	0.10594783	0.12378159	0.08940271	0.07145235	
OC =~ OC2	OC =~ OC3	OC =~ OC4	SI =~ SI1	SI =~ SI2	SI =~ SI3	SI =~ SI4	
0.03204257	0.05419105	0.04792753	0.04748498	0.04019511	0.05920661	0.04581260	

\$Loading_estimates\$bias

EP =~ EP1	EP =~ EP2	EP =~ EP3	EP =~ EP4	AC =~ AC1			
0.0119315350	-0.0188909609	0.0112732015	0.0004773045	0.0123984136			
AC =~ AC2	AC =~ AC3	AC =~ AC4	JS =~ JS1	JS =~ JS2			
0.0231852509	0.0135079989	-0.0570894809	0.0047243690	0.0012432096			
JS =~ JS3	JS =~ JS4	JS =~ JS5	OC =~ OC1	OC =~ OC2			
-0.0107473460	-0.0027082401	-0.0460121650	0.0025178269	-0.0130455845			
OC =~ OC3	OC =~ OC4	SI =~ SI1	SI =~ SI2	SI =~ SI3			
0.0034619716	-0.0037206081	0.0032970873	-0.0036331024	0.0093601379			
SI =~ SI4							
-0.0074523575							

\$Loading_estimates\$CI_standard_z

EP =~ EP1	EP =~ EP2	EP =~ EP3	EP =~ EP4	AC =~ AC1	AC =~ AC2	AC =~ AC3	
95%L	0.4542099	0.7597389	0.5787699	0.6968157	0.5932869	0.4486132	0.6572213
95%U	0.7901292	1.0354068	0.9342879	0.9483678	0.9201761	0.8580681	0.9589346
AC =~ AC4	JS =~ JS1	JS =~ JS2	JS =~ JS3	JS =~ JS4	JS =~ JS5	OC =~ OC1	
95%L	0.9598629	0.3962698	0.4839994	0.5213208	0.3920053	0.7711864	0.2826487
95%U	1.1410876	0.8439489	0.9064317	0.9366287	0.8772202	1.1216385	0.5627367
OC =~ OC2	OC =~ OC3	OC =~ OC4	SI =~ SI1	SI =~ SI2	SI =~ SI3	SI =~ SI4	
95%L	0.9077116	0.576202	0.7663514	0.7206440	0.7953715	0.5524319	0.8135955
95%U	1.0333161	0.788627	0.9542239	0.9067817	0.9529335	0.7845176	0.9931776

```

$Loading_estimates$CI_standard_t
  EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4536963 0.7593174 0.5782263 0.6964311 0.5927871 0.4479872 0.6567600
95%U 0.7906428 1.0358283 0.9348315 0.9487524 0.9206759 0.8586942 0.9593959
  AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9595858 0.3955853 0.4833535 0.5206859 0.3912634 0.7706505 0.2822204
95%U 1.1413647 0.8446334 0.9070776 0.9372637 0.8779621 1.1221744 0.5631650
  OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.9075195 0.5758772 0.7660642 0.7203594 0.7951306 0.5520771 0.8133209
95%U 1.0335082 0.7889518 0.9545111 0.9070663 0.9531744 0.7848724 0.9934521

$Loading_estimates$CI_percentile
  EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4695362 0.7141417 0.5861862 0.6910018 0.6073221 0.4551459 0.6737403
95%U 0.8058722 0.9842858 0.9390766 0.9448089 0.9418083 0.8749444 0.9751209
  AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.8258577 0.3905978 0.4727571 0.4871845 0.3683305 0.6481645 0.2770523
95%U 0.9970881 0.8437585 0.8930565 0.9021730 0.8566478 0.9895487 0.5581800
  OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.8749187 0.5830295 0.7534215 0.7238992 0.7849956 0.5645130 0.7950191
95%U 0.9959027 0.7959996 0.9427467 0.9082782 0.9439578 0.7985734 0.9756005

$Loading_estimates$CI_basic
  EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4384668 0.8108598 0.5739812 0.7003746 0.5716547 0.4317369 0.6410351
95%U 0.7748028 1.0810039 0.9268716 0.9541817 0.9061410 0.8515354 0.9424157
  AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 1.103862 0.3964602 0.4973746 0.5557765 0.4125777 0.9032762 0.2872053
95%U 1.275093 0.8496209 0.9176740 0.9707650 0.9008949 1.2446604 0.5683331
  OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.945125 0.5688295 0.7778286 0.7191474 0.8043472 0.5383761 0.8311725
95%U 1.066109 0.7817996 0.9671537 0.9035264 0.9633095 0.7724365 1.0117539

$Loading_estimates$CI_bc
  EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4395424 0.7477945 0.5522135 0.6827418 0.5665922 0.3459511 0.6321116
95%U 0.7778976 0.9956127 0.9130303 0.9391844 0.9171489 0.8240510 0.9511571
  AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9876729 0.3654902 0.4470995 0.5060276 0.3590332 0.7517272 0.2593457
95%U 0.9999848 0.8168630 0.8785544 0.9144032 0.8511615 0.9985365 0.5461919
  OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.8975169 0.5770774 0.7551351 0.7154785 0.7916389 0.5362403 0.8117293
95%U 0.9989863 0.7878350 0.9457524 0.8991749 0.9498524 0.7752105 0.9848405

$Weight_estimates
$Weight_estimates$mean
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3 AC <~ AC4
0.2455644 0.3280188 0.2958856 0.3137174 0.2759010 0.2464848 0.2947993 0.3305977
JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1 OC <~ OC2 OC <~ OC3
0.2270613 0.2520258 0.2559356 0.2270724 0.3091283 0.1754396 0.3888774 0.2836912
OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
0.3508269 0.2889617 0.3054825 0.2419354 0.3130994

```

```
$Weight_estimates$sd
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
0.02942299 0.03357839 0.02745038 0.02594824 0.03037583 0.03440725 0.02806319
AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
0.01908799 0.03915388 0.03862629 0.04024470 0.04378556 0.03528609 0.02599380
OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
0.01863876 0.02250749 0.01702047 0.01577287 0.01526872 0.01871205 0.01732339
```

```
$Weight_estimates$bias
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1
0.0030387273 -0.0080520423 0.0022231350 -0.0010831046 0.0051896131
AC <~ AC2 AC <~ AC3 AC <~ AC4 JS <~ JS1 JS <~ JS2
0.0083673086 0.0056248639 -0.0190452433 0.0047250033 0.0042029478
JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1 OC <~ OC2
0.0003667173 0.0022200357 -0.0112636922 0.0013641970 -0.0030970177
OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3
0.0029027401 0.0001601336 0.0007293655 -0.0016274889 0.0028024441
SI <~ SI4
-0.0029766061
```

```
$Weight_estimates$CI_standard_z
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1818189 0.2783105 0.2376376 0.2650260 0.2059863 0.1623132 0.2285467
95%U 0.2971549 0.4099353 0.3452411 0.3667413 0.3250573 0.2971872 0.3385524
AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3312764 0.1408711 0.1679138 0.1763240 0.1368142 0.2624962 0.1217643
95%U 0.4061000 0.2943515 0.3193261 0.3340803 0.3084504 0.4008151 0.2236581
OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3585401 0.2337718 0.3171472 0.2565887 0.2788113 0.1996555 0.2850994
95%U 0.4316027 0.3219996 0.3838662 0.3184173 0.3386636 0.2730054 0.3530058
```

```
$Weight_estimates$CI_standard_t
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1816426 0.2781092 0.2374731 0.2648705 0.2058042 0.1621070 0.2283785
95%U 0.2973313 0.4101366 0.3454056 0.3668968 0.3252394 0.2973934 0.3387206
AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3311620 0.1406365 0.1676823 0.1760828 0.1365518 0.2622847 0.1216085
95%U 0.4062144 0.2945862 0.3195576 0.3343216 0.3087129 0.4010266 0.2238139
OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3584284 0.2336369 0.3170452 0.2564942 0.2787198 0.1995434 0.2849956
95%U 0.4317144 0.3221345 0.3839682 0.3185118 0.3387551 0.2731176 0.3531097
```

```
$Weight_estimates$CI_percentile
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1862724 0.2664972 0.2383721 0.266352 0.2161097 0.1666236 0.2393982
95%U 0.3022760 0.3966996 0.3471264 0.369077 0.3392785 0.3061925 0.3497473
AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.2900372 0.1467359 0.1749539 0.1782516 0.1372702 0.2339649 0.1208154
95%U 0.3632796 0.3002947 0.3264341 0.3359858 0.3123902 0.3696986 0.2220314
OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3553286 0.2406968 0.3184753 0.2587451 0.2762347 0.2039864 0.2806058
95%U 0.4280407 0.3305418 0.3845823 0.3195894 0.3366333 0.2783462 0.3488488
```

```

$Weight_estimates$CI_basic
  EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1766979 0.2915462 0.2357523 0.2626903 0.1917651 0.1533079 0.2173518
95%U 0.2927014 0.4217486 0.3445066 0.3654153 0.3149339 0.2928768 0.3277009
  AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3740968 0.1349279 0.1608058 0.1744185 0.1328744 0.2936128 0.1233909
95%U 0.4473393 0.2884867 0.3122860 0.3321528 0.3079944 0.4293464 0.2246069
  OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3621021 0.2252296 0.3164310 0.2554166 0.2808416 0.1943147 0.2892564
95%U 0.4348143 0.3150746 0.3825381 0.3162609 0.3412402 0.2686745 0.3574994

$Weight_estimates$CI_bc
  EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1781986 0.2817962 0.2311247 0.2708809 0.2074401 0.1309097 0.2290425
95%U 0.2950485 0.4179810 0.3407272 0.3750284 0.3286914 0.2865969 0.3393959
  AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3349048 0.1322677 0.1609181 0.1775223 0.1301494 0.2548716 0.1107709
95%U 0.4126115 0.2896260 0.3139789 0.3346152 0.3051568 0.3866948 0.2161321
  OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3623654 0.2365131 0.3180438 0.2570640 0.2808688 0.1971703 0.2867658
95%U 0.4397117 0.3242966 0.3841103 0.3181764 0.3420768 0.2708492 0.3562541

$Exo_construct_correlation
$Exo_construct_correlation$mean
  EP ~~ AC
0.2608173

$Exo_construct_correlation$sd
  EP ~~ AC
0.05493887

$Exo_construct_correlation$bias
  EP ~~ AC
0.005870009

$Exo_construct_correlation$CI_standard_z
  EP ~~ AC
95%L 0.1413991
95%U 0.3567555

$Exo_construct_correlation$CI_standard_t
  EP ~~ AC
95%L 0.1410698
95%U 0.3570848

$Exo_construct_correlation$CI_percentile
  EP ~~ AC
95%L 0.1489720
95%U 0.3655552

$Exo_construct_correlation$CI_basic

```



```

      EP ~~ AC
95%L 0.1325994
95%U 0.3491826

$Exo_construct_correlation$CI_bc
      EP ~~ AC
95%L 0.1269960
95%U 0.3494058

$Indirect_effect
$Indirect_effect$mean
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
0.0255739207 -0.0008060915 0.2637702742 0.0906133368 0.0501359641

$Indirect_effect$sd
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
0.01464718 0.00685761 0.04074460 0.02905960 0.02697351

$Indirect_effect$bias
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
0.0020537265 -0.0004907846 0.0100086766 0.0066444973 0.0030606203

$Indirect_effect$CI_standard_z
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.007241482 -0.01326519 0.1638950 0.02036857 -0.008852386
95%U 0.050174418 0.01361615 0.3236109 0.13428011 0.096881833

$Indirect_effect$CI_standard_t
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.007329269 -0.01330629 0.1636508 0.02019441 -0.00901405
95%U 0.050262205 0.01365725 0.3238551 0.13445428 0.09704350

$Indirect_effect$CI_percentile
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.001161769 -0.01616362 0.1901888 0.03657312 -0.002077644
95%U 0.056265783 0.01274064 0.3471311 0.14899316 0.102420267

$Indirect_effect$CI_basic
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.01333285 -0.01238969 0.1403747 0.005655529 -0.01439082
95%U 0.04409470 0.01651457 0.2973171 0.118075566 0.09010709

$Indirect_effect$CI_bc
      OC ~ EP      OC ~ AC      SI ~ EP      SI ~ AC      SI ~ JS
95%L -0.003885354 -0.01580347 0.1758929 0.02554995 -0.008325499
95%U 0.052323822 0.01311098 0.3308011 0.13660292 0.096759834

$Total_effect
$Total_effect$mean
      JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP
0.25498139 -0.00699852 0.45974809 0.18320099 0.10066826 0.26377027

```

```

    SI ~ AC      SI ~ JS      SI ~ OC
0.09061334 0.18197253 0.49777868

$Total_effect$sd
    JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
0.05238688 0.06081747 0.05499662 0.04785428 0.05289946 0.04074460 0.02905960
    SI ~ JS      SI ~ OC
0.04375639 0.05095048

$Total_effect$bias
    JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP
0.010156430 -0.003716447 0.007446416 0.010961882 0.004598830 0.010008677
    SI ~ AC      SI ~ JS      SI ~ OC
0.006644497 0.003670602 0.007764904

$Total_effect$CI_standard_z
    JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
95%L 0.1319921 -0.1187657 0.3370639 0.06748455 -0.01221044 0.1638950 0.02036857
95%U 0.3373449 0.1196344 0.5526467 0.25506989 0.19515164 0.3236109 0.13428011
    SI ~ JS      SI ~ OC
95%L 0.08887038 0.3823878
95%U 0.26039228 0.5821100

$Total_effect$CI_standard_t
    JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
95%L 0.1316782 -0.1191302 0.3367343 0.06719774 -0.01252749 0.1636508 0.02019441
95%U 0.3376589 0.1199989 0.5529763 0.25535671 0.19546869 0.3238551 0.13445428
    SI ~ JS      SI ~ OC
95%L 0.08860813 0.3820824
95%U 0.26065453 0.5824154

$Total_effect$CI_percentile
    JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
95%L 0.1525445 -0.1247503 0.3533574 0.090650 -0.004465693 0.1901888 0.03657312
95%U 0.3584825 0.1147835 0.5683438 0.272157 0.200425662 0.3471311 0.14899316
    SI ~ JS      SI ~ OC
95%L 0.09737975 0.3987664
95%U 0.26719106 0.5977819

$Total_effect$CI_basic
    JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP
95%L 0.1108546 -0.1139147 0.3213667 0.05039744 -0.01748446 0.1403747
95%U 0.3167926 0.1256190 0.5363531 0.23190445 0.18740689 0.2973171
    SI ~ AC      SI ~ JS      SI ~ OC
95%L 0.005655529 0.0820716 0.3667158
95%U 0.118075566 0.2518829 0.5657313

$Total_effect$CI_bc
    JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP      SI ~ AC
95%L 0.1272011 -0.1176517 0.3393091 0.06572548 -0.01591844 0.1758929 0.02554995
95%U 0.3350133 0.1223213 0.5556759 0.25411065 0.18937667 0.3308011 0.13660292
    SI ~ JS      SI ~ OC
95%L 0.09090796 0.3873681

```

95%U 0.26100956 0.5859620

```
attr("class")
[1] "cSEMInfer"
```

6. Model fit ----

```
testOMF(res, .R = 10000, .seed = 42,
        .handle_inadmissibles = "replace")
```

----- Test for overall model fit based on Beran & Srivastava (1985) -----

Null hypothesis:

```
+-----+
|
|  H0: The model-implied indicator covariance matrix equals the  |
|  population indicator covariance matrix.                      |
|
+-----+
```

Test statistic and critical value:

Distance measure	Test statistic	Critical value
		95%
dG	0.3253	0.3554
SRMR	0.0706	0.0505
dL	1.1505	0.5898
dML	1.8972	2.1208

Decision:

Distance measure	Significance level
	95%
dG	Do not reject
SRMR	reject
dL	reject
dML	Do not reject

Additional information:

Out of 19853 bootstrap replications 10000 are admissible.
See ?verify() for what constitutes an inadmissible result.

The seed used was: 42

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